

Bias and Representation

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of representation and bias evaluation.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 30: Bias and Representation

Outputs should be tested for omission, stereotyping, and uneven quality.



CREATOR MISSION

Audit a character generator across varied prompts.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Copyright and permitted sources. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain bias and representation using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use bias and representation to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Bias evaluation tests defined outcomes across relevant conditions. Group averages can expose differences but cannot alone establish their cause or the correct remedy.

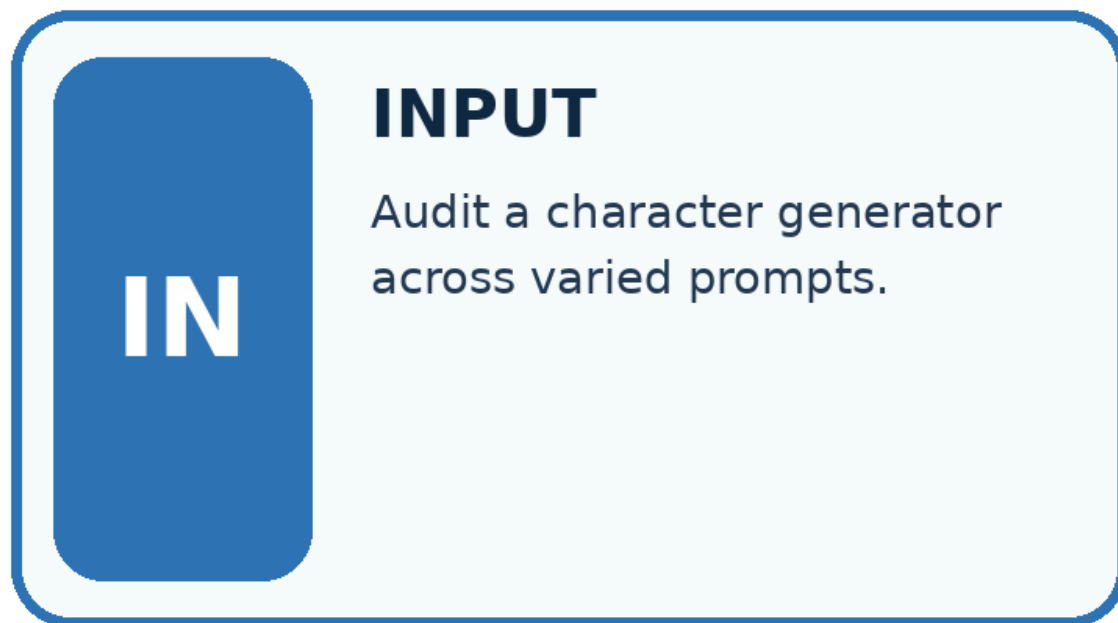
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Outputs should be tested for omission, stereotyping, and uneven quality.



OUT

EVIDENCE

Audit a character generator across varied prompts.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Representation and bias evaluation — evidence trace

Input and scope · **Check or calculation** · **Conclusion and limit**

Input and scope: A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases.

Check or calculation: Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Conclusion and limit: Rates 90% and 60%; gap 30 percentage points. Investigate case mix, retrieval coverage and annotation rules before attributing cause.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Pine instead has 6 supported answers out of 12.

Prediction: Pine is 50%, which is 25 percentage points below Birch.

Reason: Equal correct counts do not imply equal rates when denominators differ.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

REPRESENTATION

who and what appears in
outputs

2

STEREOTYPE

oversimplified assumption
about a group

3

SUBGROUP QUALITY

performance for a
meaningful group

4

AUDIT

systematic evidence check

Word · My own example · Not this / boundary

Word: representation

My own example: A way to encode information so a person or system can work with it.

Not this / boundary: _____

Word: stereotype

My own example: An oversimplified expectation about members of a group that ignores individual variation.

Not this / boundary: _____

Word: subgroup quality

My own example: Performance measured for a defined subset using stated criteria and denominators.

Not this / boundary: _____

Word: audit

My own example: A systematic examination of records and behaviour against stated checks.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Subgroup rate

Meaning: A measured outcome within a stated group

Word: Percentage-point gap

Meaning: The difference between two percentage rates

Word: Case mix

Meaning: The distribution of task conditions in a group

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 30 • WORKED EVIDENCE

- 1 Generate comparable characters while changing one identity descriptor.
- 2 Count omissions, stereotyped roles, and quality differences.
- 3 A balanced count does not alone prove respectful depiction.
- 4 Repair prompt, data, filters, or use limits, then repeat the fixed audit.

1. Generate comparable characters while changing one identity descriptor.
2. Count omissions, stereotyped roles, and quality differences.
3. A balanced count does not alone prove respectful depiction.

4. Repair prompt, data, filters, or use limits, then repeat the fixed audit.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases.

Rates 90% and 60%; gap 30 percentage points. Investigate case mix, retrieval coverage and annotation rules before attributing cause.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Bias and Representation

Compare study-answer support rates when groups have different numbers of questions.

The facts we will use

Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases. Both groups use the same support rubric.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Bias and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep denominators

Birch: 6/8. Pine: 9/12.



2. Compare rates

Both support rates are 75%; Pine has more correct answers because it has more cases.



3. Inspect errors

Check the two Birch and three Pine failures before deciding what revision each needs.

CHANGE ONE THING

Pine instead has 6 supported answers out of 12.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Keep denominators

What happens and why: Birch: 6/8. Pine: 9/12.

Step: 2. Compare rates

What happens and why: Both support rates are 75%; Pine has more correct answers because it has more cases.

Step: 3. Inspect errors

What happens and why: Check the two Birch and three Pine failures before deciding what revision each needs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

Read the labelled picture: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Pine instead has 6 supported answers out of 12. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Fairness is solved when every group appears equally often. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help

used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal mechanism.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Representation and bias evaluation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Representation and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Create paired prompts, score quality and stereotypes, compare patterns, and propose one controlled mitigation with retest.

Create paired prompts, score quality and stereotypes, compare patterns, and propose one controlled mitigation with retest.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Audit answer quality across reading levels, languages, names, and accessibility needs.

THIS WEEK’S CHECKPOINT Audit answer quality across reading levels, languages, names, and accessibility needs.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Audit a character generator across varied prompts.

CHALLENGE Audit a character generator across varied prompts.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Scientist images repeatedly show only one demographic group.

The question

Evaluate and improve representation.

Worked explanation

Review relevant representation criteria across outputs, revise inclusive requirements and retest. Appearance alone cannot establish absence of all bias.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Review representation across outputs

Evidence or stage · Value or result

Evidence or stage: Observed pattern

Value or result: Only one demographic group shown

Evidence or stage: Review

Value or result: Define relevant representation criteria

Evidence or stage: Revision

Value or result: State inclusive requirements

Evidence or stage: Retest

Value or result: Inspect multiple new outputs

Scientist images repeatedly show only one demographic group.

Evaluate and improve representation.

Review relevant representation criteria across outputs, revise inclusive requirements and retest.

Appearance alone cannot establish absence of all bias.

14. Try a new case independently

Independent case: Apply a new case independently

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why

raw correct totals mislead.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An assistant is tested on six fluent requests but no requests with spelling errors. What gap does this leave, and how should the evaluation expand?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Audit answer quality across reading levels, languages, names, and accessibility needs.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

An illustration set always shows scientists in the same role and appearance. Define a review that examines representation without inferring personal identities from images.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Compare study-answer support rates when groups have different numbers of questions.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of representation and bias evaluation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of representation and bias evaluation

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.